FORECASTING / PRODUCTION MLOPS

From a forecasting notebook to repeatable planning inputs

Make forecasts available through an API, automate repeatable workflows and keep model changes controlled.

WHY MOVE BEYOND THE NOTEBOOK?

Reliable delivery

Predictions for applications, without manual notebook runs and file exports.

Less manual work

Repeatable checks, training and releases with traceable results.

Controlled updates

Monitor quality, evaluate candidates and verify the rollback path.

7.5% lower RMSE

Final simulation RMSE: 1099.00 → 1016.51.

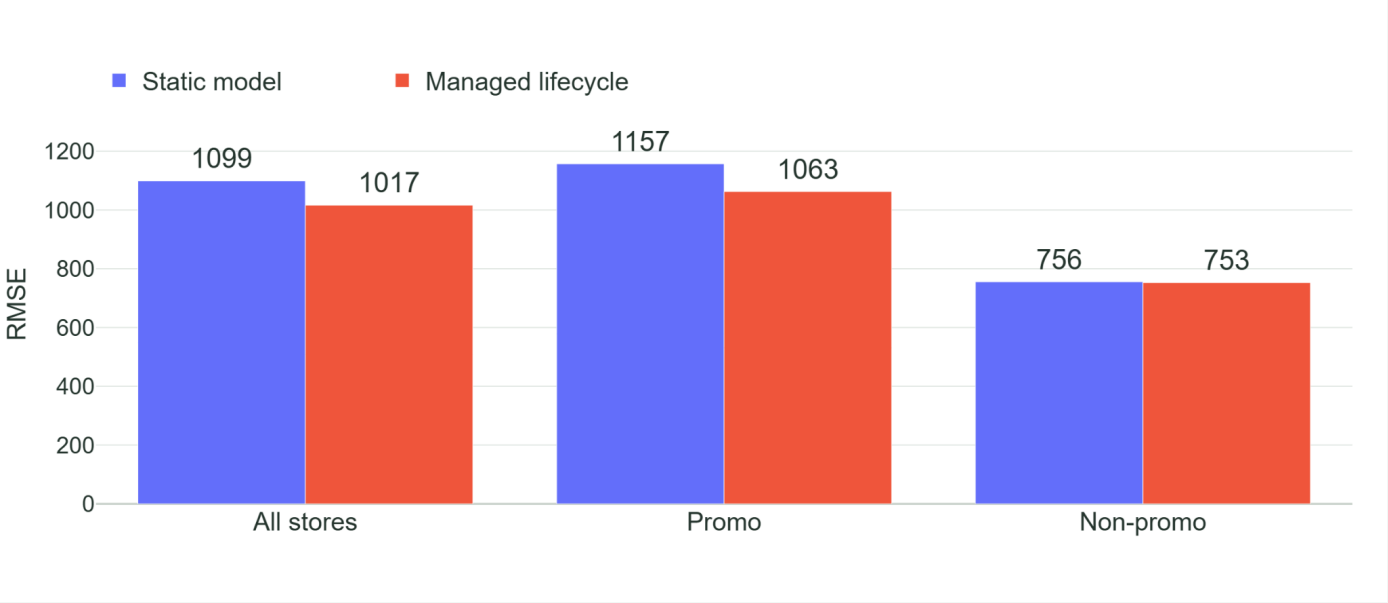
1 of 3 promoted

Three retraining events; one model promotion.

Rollback verified

100% traffic restored; readiness HTTP 200.

MODEL QUALITY BY STORE SEGMENT



Controlled drift simulation. Lower RMSE is better; the main improvement occurs in promo stores.

TESTED RECOVERY AFTER AN APPLICATION UPDATE

Cloud Run rollback

- Environment: `dev`
- Service: `mlops-sales-forecasting-dev-api`
- Revision: `mlops-sales-forecasting-dev-api-00001-brz`
- Traffic: `100%`
- Readiness: `HTTP 200`

Built with FastAPI, Prefect, MLflow, Docker, GitHub Actions, Terraform, Cloud Run and Grafana.

Independent reference project using public Rossmann data. Results are from a controlled simulation, not a client deployment. Cloud deployment and rollback successfully verified.

Have a working model? Start with a Notebook-to-MLOps Assessment.

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